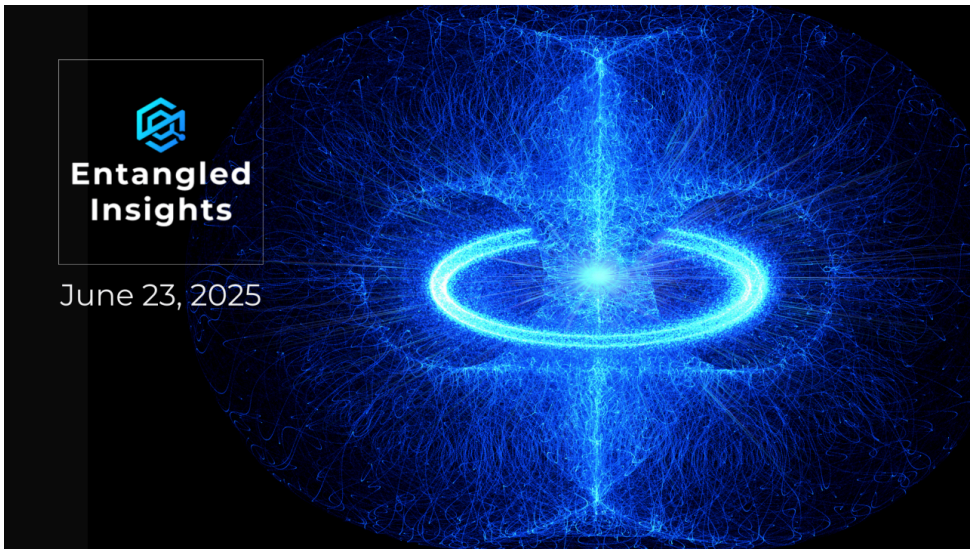
Entangled Insights



Entangled Insights #6: Inside IBM's Starling Leap, NIST-Level Trust, and the Dark Side of the Web

SQE.io

June 23, 2025

📬 **Welcome back to *Entangled Insights***

Cybersecurity is evolving faster than most teams can track. This edition is all about momentum. From major quantum breakthroughs to hands-on masterclasses and a closer look at what it takes to implement Zero Trust the right way, this issue cuts through the noise to bring you what matters now.

In this issue:

- A closer look at IBM's 2029 quantum ambitions
- What makes Quantum Foundry's work stand out
- Why SQE isn't just aligned with NIST but built to deliver
- The podcast fueling our thinking this month

Let's get started!

Partner Spotlight: Quantum Foundry



Quantum Foundry equips enterprises with future-proof solutions that protect critical infrastructure and guide strategic innovation. Their expertise lies in two essential areas:

Quantum Security Solutions

Enterprises face increasing pressure to prepare for post-quantum threats. Quantum Foundry provides:

- Quantum-safe networks
- Encrypted communications
- Quantum-resistant key distribution
- Secure data encryption

These tools are designed to defend core systems against the looming risks posed by quantum computing.

AI Data Center Readiness

Most organizations struggle to assess where they stand with AI and quantum readiness. Quantum Foundry solves that by:

- Evaluating your current security and AI posture
- Identifying gaps and vulnerabilities
- Delivering a roadmap for safe, scalable adoption

Their approach helps decision-makers plan with confidence in a rapidly evolving tech landscape.

> *"By combining* **Quantum Foundry***'s cutting-edge quantum hardware with* ***SQE.io****'s quantum-secure blockchain, we provide a holistic and unhackable foundation for critical enterprise data and communication."* — **Bill Xydias**, VP of Marketing Innovation

Together, SQE and Quantum Foundry offer real protection for real-world systems, building infrastructure that holds strong in the quantum era.

Learn more about Quantum Foundry at: www.quantumfoundry.ai

Speaking of Quantum Foundry... see you in Indianapolis!



If you're near Indianapolis this July, save the date. Quantum Foundry is hosting a Quantum Safe Networking Masterclass, and it's the kind of event that cybersecurity professionals will not want to miss.

- **Thursday, July 17**
- **16 Tech | 1220 Waterway Boulevard | Indianapolis, IN**
- **1 to 5 PM EST**
- **Happy hour to follow**

This hands-on masterclass is built for people who are ready to move beyond buzzwords and get into the real strategy behind quantum-safe networks. It will be led by Sylvain Chenard, ing., Nokia's lead for Quantum-Safe Network Solutions. With nearly three decades of

experience, Sylvain will walk through the real risks, what's coming next, and how to build networks that stand up to quantum-level threats.

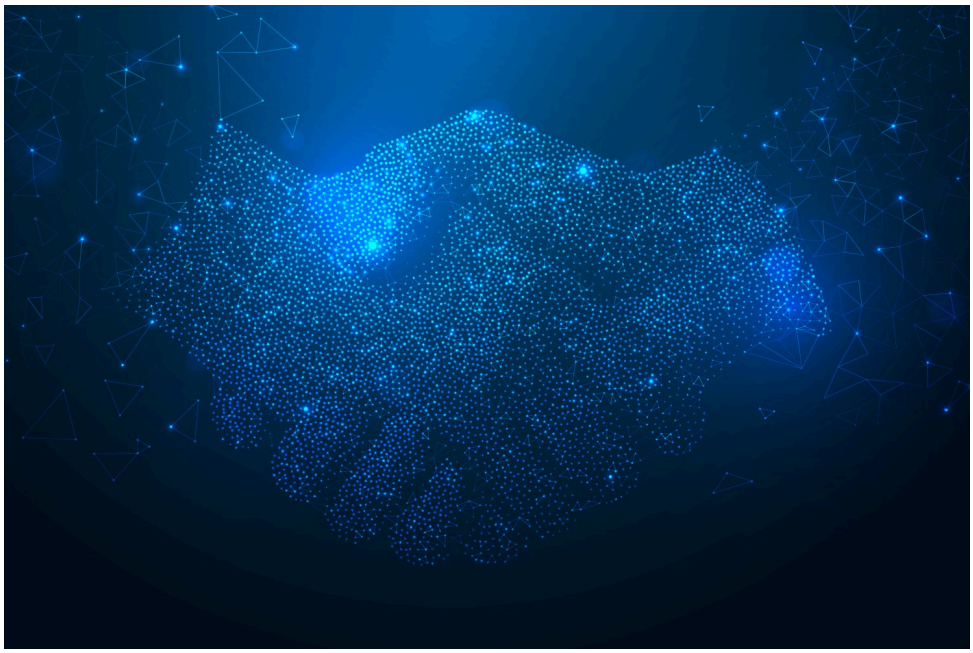
You'll learn:

- What the quantum threat means for secure communications today
- How entropy, keys, and encryption work together in a post-quantum framework
- Key distribution methods including public key cryptography, pre-shared keys, and quantum key distribution
- Crypto-agility and defense-in-depth strategies for enterprise protection
- How to apply these principles directly to your infrastructure

This session is ideal for anyone responsible for network security, encryption strategy, or long-term infrastructure planning.

Register or learn more here.

SQE: Built for Zero Trust From the Ground Up



National Institute of Standards and Technology (NIST)'s latest update to its Zero Trust Architecture framework reinforces what security leaders have long known. Trust must be earned continuously, and access should be granted only with verified identity and contextual awareness. The challenge is execution, especially when teams are forced to rely on fragmented tools and legacy infrastructure.

SQE was built to solve that.

SQE offers a unified Zero Trust platform where encryption, identity, access, automation, and monitoring are integrated from the start. At the center of it all is the **SQid**, a 1024-bit cryptographic identifier that ties every entity to a unique and verifiable identity. This isn't theoretical alignment with NIST standards. This is Zero Trust in action.







Identity At The Core

SQE's identity framework authenticates every user, device, and application with a SQid. This enables true least-privilege enforcement. Only authorized **SQids** can interact, and those relationships are controlled at the protocol level.

Quantum-Secure Communication

QueSAFE's Virtual Quantum Network (VQN) provides a modern approach to remote access and data protection. Instead of traditional VPNs, it creates direct, encrypted tunnels between validated endpoints. SQE has no visibility into the encrypted content, preserving both privacy and compliance.

Real-Time Visibility And Automated Enforcement

The SQE Monitor gives your team a read-only view into all infrastructure and application activity. Smart contracts manage permissions and system changes, reducing human error while enforcing policy across CI/CD, databases, and AI systems.

Secure Automation Across Environments

From manufacturing to AI training, SQE's **no-code smart contract** engine ensures that every action is verified and logged. If a step falls out of compliance, it is flagged or stopped automatically.

Verified Endpoint Compliance

With built-in Know-Your-Data and KYD for IoT, SQE authenticates communication at the human and machine level. If a device or endpoint fails compliance, it cannot access resources or participate in transactions.

Encrypted, Structured Data Management

Queverse, SQE's vectorized database built on distributed S3 architecture, enables fast, encrypted storage across a 3D file space. The system supports high-speed search, secure access, and serves as the foundation for future AR integration.

National Institute of Standards and Technology (NIST) has provided the blueprint. SQE delivers a complete environment that meets the requirements without compromising scalability or performance.

Walk through SQE's core features and discover how identity-bound automation keeps your data protected at **SQE.io**

What We're Listening To: Darknet Diaries



Darknet Diaries – True stories from the dark side of the Internet.

Imagine driving down the road and your phone loses signal. Not just a lost signal, but it starts displaying SOS. Meaning no service configured a...

darknetdiaries.com

We've been hooked on *Darknet Diaries* lately! Hosted by **Jack Rhysider**, this podcast dives into the real stories behind cybercrime, data breaches, hacktivism, and the strange but true moments that shape our digital world.

From the bizarre rise of fake hitman-for-hire sites, to the dramatic crypto theft that made headlines, to insider stories from DEF CON legends, every episode offers a reminder of why strong encryption, verifiable identity, and zero trust infrastructure matter more than ever.

It's one of the few shows that keeps us entertained while reinforcing the urgency of what we're building at SQE.

🔗 **Listen here** and let us know which episode got you thinking!

The Quantum Clock Is Ticking



IBM just announced its roadmap for a fault-tolerant quantum computer, and the timeline might be shorter than many expected. Set for 2029, their new system, Quantum Starling, is being designed to run 100 million quantum operations using 200 error-corrected qubits, a major step forward in quantum computing stability and scale.

Fault tolerance matters because today's quantum systems still struggle with high error rates. IBM's new approach uses Bivariate Bicycle codes to drastically reduce how many physical qubits are needed. If successful, this could move the industry closer to real-world applications like molecular modeling and, eventually, post-quantum cryptographic threats.

The implications go beyond blockchain. From banking systems to cloud services, stronger quantum computing could put pressure on legacy encryption models across the board.

Read the full breakdown here

Thanks for reading *Entangled Insights*! We'll return in a few weeks with more updates, signals, and emerging shifts in cybersecurity and quantum tech.

SQE: Your World, Quantum Secure.

📥 Download the QueSAFE Product Sheet

💬 Book a Demo

🔎 Explore Our Technology

🛠️ Developer Pre-Registration

📱 SQE App Pre-Registration

▶️ SQE Youtube

About SQE

SQE is redefining cybersecurity with a quantum-secure platform built for the future. Built on simulated quantum entanglement and decentralized hardware, the SQE platform eliminates key transmission, enabling scalable, tamper-proof security across industries.